

Measurement-updated Stokes prediction and reduced-visibility actuation test

Hybrid MZI polarization-control system: measured-state model validation, improved orthogonality, and Red Pitaya V_π/V_λ command-chain verification

Scope and data philosophy. This report uses measured optical powers and measured Stokes states as inputs to the model. No polarization state, overlap, or visibility parameter is adjusted to force agreement with a measured fringe. The model prediction is therefore forward-propagated from independently measured quantities. Visibility estimates from swept data are reported with the same extrema/envelope logic used in the earlier report so that the comparison is traceable across alignment states.

1. Baseline result: measurement-updated Jones/Stokes visibility model

1.1 Minimal update to the ideal model

The ideal output field at the monitored recombination port is taken to contain two coherent contributions whose relative phase is controlled by ϕ_1 . Replacing ideal orthogonal basis states by the *measured* normalized polarization states $|A\rangle$ and $|B\rangle$ gives

$$E_4(\phi_1) = \sqrt{P_A} e^{i\phi_1} |A\rangle + \sqrt{P_B} |B\rangle. \quad (1)$$

The corresponding intensity is

$$I_4(\phi_1) = P_A + P_B + 2\sqrt{P_A P_B} \Re[e^{i\phi_1} \langle B|A \rangle]. \quad (2)$$

Hence the theoretical fringe visibility is

$$V_{\text{th}} = \frac{2\sqrt{P_A P_B}}{P_A + P_B} |\langle A|B \rangle|. \quad (3)$$

For fully polarized states, the Jones-state overlap follows directly from the normalized Stokes vectors,

$$|\langle A|B \rangle| = \sqrt{\frac{1 + \mathbf{s}_A \cdot \mathbf{s}_B}{2}}. \quad (4)$$

Equation (??) separates two physically distinct contrast terms: a scalar power-balance factor and a polarization-overlap factor.

1.2 Earlier measured-state substitution

The earlier isolated-state measurement used

$$\begin{aligned} \mathbf{s}_A &= (+0.990466, -0.051196, +0.127887), & P_A &= 0.342\,329 \text{ mW}, \\ \mathbf{s}_B &= (-0.989706, -0.092665, -0.109066), & P_B &= 0.417\,919 \text{ mW}. \end{aligned}$$

Therefore

$$\begin{aligned} \mathbf{s}_A \cdot \mathbf{s}_B &= -0.989475, \\ |\langle A|B \rangle| &= \sqrt{\frac{1 - 0.989475}{2}} = 0.072544, \\ \frac{2\sqrt{P_A P_B}}{P_A + P_B} &= 0.995045, \end{aligned}$$

which yields

$$V_{\text{th},0} = (0.995045)(0.072544) = 0.07219 \approx \mathbf{7.22\%}. \quad (5)$$

The corresponding swept-PAX result was

$$P_{\text{min}} = 0.7264 \text{ mW}, \quad P_{\text{max}} = 0.8615 \text{ mW},$$

with a literal-extrema visibility of 8.51%, a repeated-cycle mean of $6.60\% \pm 0.61\%$, and a pooled 2–98% envelope visibility of 7.23%.

Baseline quantity	Visibility	Interpretation
Stokes/Jones prediction	7.22%	Independent isolated-state inputs
PAX pooled 2–98% envelope	7.23%	Robust repeated-sweep estimate
PAX cycle-wise mean	6.60% $\pm$ 0.61%	Sparse sampling lowers extrema
PAX literal global extrema	8.51%	Most sensitive to single samples

Baseline milestone. The independently measured Stokes overlap predicted 7.22% visibility, while the robust swept-PAX estimate was 7.23%. The earlier residual fringe contrast was therefore quantitatively explained by measured polarization nonorthogonality without fitting the model to the visibility data.

2. Improved orthogonalization: updated measured states and forward visibility prediction

2.1 Time-averaged isolated-path measurements

The newer isolated-path acquisition contains 60 s records for `path_a_only` and `path_b_only`. We identify these with the two coherent recombination contributions E_1 and E_2 and retain the A/B notation of Section 1. The Stokes directions and powers below are arithmetic time averages of the recorded values; they are not optimization variables.

Quantity	Path A (E_1)	Path B (E_2)
$\langle S_1 \rangle$	+0.989929	−0.989767
$\langle S_2 \rangle$	−0.056929	+0.050230
$\langle S_3 \rangle$	+0.129615	−0.133559
$\langle P \rangle$	0.346 664 mW	0.377 278 mW
σ_P	0.000 325 mW	0.000 417 mW
Mean reported DOP	1.01097	0.99899

The small $\text{DOP} > 1$ excursion in path A is an instrument/calibration artifact; the recorded Stokes direction itself has unit norm to numerical precision and is used only as a polarization direction in Eq. (??).

Using the measured mean vectors,

$$\mathbf{s}_A \cdot \mathbf{s}_B = -0.99996975, \quad (6)$$

$$|\langle A|B \rangle| = \sqrt{\frac{1 - 0.99996975}{2}} = 0.003889, \quad (7)$$

$$\frac{2\sqrt{P_A P_B}}{P_A + P_B} = 0.999105. \quad (8)$$

Therefore the forward prediction from the re-aligned isolated measurements is

$$V_{\text{th},1} = (0.999105)(0.003889) = 0.003886 \approx \mathbf{0.389\%}. \quad (9)$$

The power-balance term is essentially unity. In this alignment, the model therefore predicts that the residual fringe floor should be set almost entirely by the remaining polarization overlap.

2.2 Comparison against the subsequent swept measurement

The subsequent command-chain acquisition sweeps the phase actuator continuously over one calibrated full-wave command range. To preserve the reduction routine of the baseline report, the first 1 s is treated as settling and three direct visibility estimators are reported from the remaining PAX total-power samples:

$$V = \frac{P_{\text{max}} - P_{\text{min}}}{P_{\text{max}} + P_{\text{min}}}. \quad (10)$$

For the updated sweep,

Updated quantity	Visibility	Interpretation
Stokes/Jones prediction	0.389%	Independent time-averaged E_1/E_2 states and powers
PAX pooled 2–98% envelope	3.04%	Robust full-record envelope after settling
PAX cycle-wise mean	$2.68\% \pm 0.64\%$	27 complete/reliably sampled drive cycles
PAX literal global extrema	4.96%	Most sensitive to isolated power samples

Relative to the earlier robust envelope value of 7.23%, the updated 2–98% envelope is reduced to 3.04% (approximately a 58% reduction). The cycle-wise estimator similarly falls from 6.60% to 2.68% (approximately a 59% reduction).

Updated visibility result. Orthogonalizing the isolated states substantially reduces the measured contrast, in the direction predicted by the Stokes/Jones model. However, the simple two-state model now predicts 0.389%, below the observed ~ 2.7 – 3.0% robust contrast range. The remaining contrast should therefore be treated as an unresolved experimental floor rather than absorbed into a fitted polarization overlap. Candidate contributions include residual time-dependent state motion, path-power drift, actuator-correlated beam motion, finite PAX temporal averaging, or other non-ideal interferometer terms not represented by the two static Jones states.

3. Symbolic triage of the V_π/V_λ Red Pitaya command chain

3.1 Command definition and measured electrical loopback

The final CSV records a sinusoidal Red Pitaya output with

$$f_d = 0.465\,661\,287\,3\,\text{Hz}, \quad (11)$$

$$V_{\text{amp}} = 0.359\,985\,351\,6\,\text{V}, \quad (12)$$

$$V_{\text{off}} = 0.359\,985\,351\,6\,\text{V}. \quad (13)$$

Thus the command is

$$V_{\text{RP}}(t) = V_{\text{off}} + V_{\text{amp}} \sin(2\pi f_d t), \quad (14)$$

with the calibrated command-side identification

$$\boxed{V_\pi = 0.359\,985\,\text{V}, \quad V_\lambda = 2V_\pi = 0.719\,971\,\text{V}.} \quad (15)$$

These are *Red-Pitaya-output-coordinate* voltages. Any downstream high-voltage gain is intentionally not inferred from this dataset.

The recorded input loopback verifies the electrical command chain directly. Across the full record,

$$\begin{aligned} V_{\text{cmd},\text{min}} &\approx 0.000\,042\,\text{V}, & V_{\text{cmd},\text{max}} &\approx 0.719\,852\,\text{V}, \\ V_{\text{in},\text{min}} &\approx -0.004\,940\,\text{V}, & V_{\text{in},\text{max}} &\approx 0.721\,088\,\text{V}. \end{aligned}$$

A linear comparison of recorded loopback versus commanded voltage gives a slope of 1.00808, offset of $-4.93\,\text{mV}$, and correlation coefficient $r = 0.999994$. The RMS command-loopback difference is approximately $3.03\,\text{mV}$. This establishes that the commanded $0 \rightarrow V_\lambda$ excursion is physically present at the monitored RP electrical interface.

3.2 Phase mapping to the Poincaré sphere

With V_λ defined as the command producing a 2π relative phase excursion, the commanded optical phase is

$$\boxed{\phi(V) = 2\pi \frac{V}{V_\lambda}.} \quad (16)$$

Therefore

$$V = 0 \Rightarrow \phi = 0, \quad V = V_\pi = \frac{V_\lambda}{2} \Rightarrow \phi = \pi, \quad V = V_\lambda \Rightarrow \phi = 2\pi. \quad (17)$$

For a nearly balanced superposition of nearly orthogonal basis states, sweeping ϕ primarily rotates the Stokes direction around the S_2 – S_3 plane. In the ideal limiting case,

$$\mathbf{s}(\phi) = \begin{pmatrix} 0 \\ \cos(\phi + \phi_0) \\ \sin(\phi + \phi_0) \end{pmatrix}, \quad (18)$$

so a V_π command produces an antipodal point,

$$\mathbf{s}(\phi + \pi) = -\mathbf{s}(\phi), \quad (19)$$

whereas a V_λ command returns the state,

$$\mathbf{s}(\phi + 2\pi) = \mathbf{s}(\phi). \quad (20)$$

The PAX request/receive timestamps show a median acquisition-channel latency of approximately 144.3 ms (mean 144.7 ms). Evaluating the known RP sine command at the recorded PAX request time, rather than the later row-completion time, provides a direct clock registration without altering any optical parameter. Using normalized averages of samples within ± 10 mV of the three calibrated command levels gives

$$\begin{aligned} \hat{\mathbf{s}}_0 &= (-0.1852, -0.3924, +0.9010), \\ \hat{\mathbf{s}}_\pi &= (+0.1033, +0.3311, -0.9379), \\ \hat{\mathbf{s}}_\lambda &= (-0.1841, -0.2349, +0.9544). \end{aligned}$$

The corresponding dot products are

$$\boxed{\hat{\mathbf{s}}_0 \cdot \hat{\mathbf{s}}_\pi = -0.9941, \quad \hat{\mathbf{s}}_0 \cdot \hat{\mathbf{s}}_\lambda = +0.9862.} \quad (21)$$

Thus the measured V_π state is nearly antipodal to the zero-command state, while the measured V_λ state nearly returns to it. This is the expected topological behavior of a calibrated π and 2π relative-phase command on the Poincaré sphere.

3.3 Contrast observed during the calibrated full-wave actuation

The same sweep provides a reduced-contrast measurement while the Stokes state traverses the sphere. After the first 1 s settling region, the directly observed total-power visibility is summarized by

$$\begin{aligned} V_{2-98} &= 3.04\%, \\ \langle V_{\text{cycle}} \rangle &= 2.68\% \pm 0.64\%, \\ V_{\text{extrema}} &= 4.96\%. \end{aligned}$$

These values are empirical reductions of the measured PAX power record. They are not used to back-solve or tune the Stokes overlap in Eq. (??).

4. Interpretation and next diagnostic step

The two milestones together establish a useful separation between *model validation* and *residual-system diagnosis*:

1. The earlier alignment provided a close quantitative validation of the measurement-updated Jones/Stokes visibility model: 7.22% predicted versus 7.23% robustly measured.
2. The improved alignment produces much more nearly antipodal isolated Stokes states and lowers the measured fringe contrast by roughly 60% using like-for-like reduction metrics.
3. The updated static-state model predicts only 0.389% visibility, while the calibrated sweep retains approximately 2.7–3.0% robust contrast. The appropriate scientific conclusion is therefore not to modify the measured Stokes parameters, but to identify which additional time-dependent or non-Jones contribution sets the present contrast floor.
4. Independently, the RP command telemetry and Poincaré-sphere response verify the calibrated V_π/V_λ phase-control chain: V_π produces a nearly antipodal polarization state and V_λ nearly closes the trajectory.

Milestone statement. The system has progressed from a regime in which measured static polarization nonorthogonality quantitatively explained the fringe visibility to a lower-contrast regime in which the same independently measured model predicts a substantially smaller residual than is observed. At the same time, the calibrated V_π/V_λ actuator command is demonstrably mapped onto the expected Poincaré-sphere motion. The next model refinement should therefore target dynamic/systematic contrast terms, while preserving the independently measured static Stokes states as fixed experimental inputs.

5. Data provenance and reproducibility notes

- Baseline symbolic structure and first milestone values: `stokes_visibility_comparison(1).tex`.
- Updated isolated-state record: `data(20260924-004241).csv`. Each of `path_a_only`, `path_b_only`, and `both_paths` contains 241 samples spanning approximately 60 s.
- Calibrated actuation record: `data(20260924-004258).csv`, 349 samples spanning approximately 60 s.
- Static-model quantities use arithmetic time averages of independently recorded isolated-path Stokes vectors and PAX powers. No visibility datum is used to determine $\langle A|B \rangle$.
- Sweep visibility uses the same direct definition $V = (P_{\max} - P_{\min}) / (P_{\max} + P_{\min})$ as the baseline report. The pooled value uses the 2nd and 98th percentiles; the cycle statistic is the mean and standard deviation of per-cycle extrema after the settling interval.
- The V_π/V_λ state comparison uses the recorded PAX request timestamp to evaluate the already-known sinusoidal command, thereby accounting for acquisition latency without fitting an optical response parameter.